
Payments & PhonePe: Dual Payment Architecture (Wallet + Instant Gateway)

A plain-English guide to how payments work in V FOODS, why we support both a **Prepaid Campus Wallet** and an **Instant Payment Gateway (PhonePe / Paytm UPI)**, and how transactions remain fast and secure.

The Dual Payment Choice: Flexibility & Speed

V FOODS provides two flexible payment methods directly on the checkout screen:

1. **Option A: V FOODS Campus Wallet (Prepaid 1-Tap Checkout)**
 2. **How it works:** Students add money to their digital campus wallet when convenient (dorm, home, or morning) via PhonePe / UPI.
 3. **Why use it:** When buying food during a busy 10-minute break, deducting from the prepaid balance takes under one second with zero bank redirects or OTP delays.
 4. **Top-Up:** Recharge anytime in ₹100, ₹200, ₹500 increments or top up the exact deficit needed for an order.
 5. **Option B: Instant Payment Gateway (PhonePe / Paytm UPI Direct Checkout)**
 6. **How it works:** Students can pay directly per order using PhonePe, Paytm, Google Pay, or any UPI app.
 7. **Why use it:** Perfect for students ordering for the first time, visitors, or anyone who prefers direct bank debit per transaction without maintaining a prepaid wallet float.
 8. **Zero Prior Top-Up:** Places the order immediately upon bank approval without touching the wallet balance.
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The Exact Payment Journey

Here is the exact step-by-step path money takes through V FOODS:

Step 1: Student Initiates a Wallet Top-Up

- The student opens the **Wallet** tab, chooses an amount (such as ₹200), and selects their payment method (UPI or Card).
- **Fee Rules:**
- **UPI / RuPay (0% Fee):** If a student adds ₹200 via UPI, they pay exactly ₹200 and receive ₹200 in their wallet.
- **Bank Cards (Reverse Fee Pass-Through):** Credit and debit cards carry a standard bank processing fee. The app uses the reverse formula ($\text{Amount} / 0.9765$) to charge the small extra fee to the card, so the student's wallet still receives the clean ₹200 requested.

Step 2: PhonePe Collects the Payment

- V FOODS creates a secure transaction payload and passes the student to PhonePe.
- The student authorizes the payment using their preferred UPI app (PhonePe, Google Pay, BHIM) or bank card.

Step 3: PhonePe Sends a Secure Webhook Confirmation

- The moment the money leaves the student's bank account, PhonePe sends an automated notification called a **webhook** directly to our FastAPI server.
- The notification includes the transaction ID, amount paid, and a cryptographic digital signature (`X-VERIFY`).

Step 4: Server Verifies the Signature

- Before trusting the message, our server recalculates the signature using our private secret key (`SALT_KEY`).
- If the signature matches, the server knows the message genuinely came from PhonePe and was not tampered with.

Step 5: Wallet Is Credited Idempotently

- Our server calls the `credit_wallet()` stored procedure in Supabase.
- The database adds the money to the student's wallet and writes an unchangeable audit log in `wallet_txns` .
- **Double-Credit Protection:** The transaction ID is locked. If PhonePe's server sends the confirmation twice by mistake, the database detects the duplicate and ignores it, ensuring the student is never credited twice.

Step 6: Instant Food Ordering

- Later, when the student wants to eat, they select their food and tap **Place Order**.
- The database checks their wallet balance and deducts the meal cost instantly.
- The kitchen screen chimes and cooks start preparing the food.
- **PhonePe is not involved at checkout:** No bank connections, no OTPs, and zero loading spinners.

Why Checkout Doesn't Wait on PhonePe

Separating wallet top-up from food ordering is the key design choice that makes V FOODS fast:

Typical Food App (Gateway at Checkout)	V FOODS (Wallet-First Model)
Gateway opens during lunch rush	Gateway used before lunch at leisure
Requires strong cellular signal at counter	Requires zero internet speed at counter
30 to 60 second bank delay per order	Under 1 second instant database deduction
Payment failures leave students hungry in line	Zero checkout payment failures
Counter cashiers wait on banking screens	Counter cashiers hand over pre-paid meals

How the App Knows a Payment is Genuine

How do we stop someone from pretending they paid ₹500 when they didn't?

1. **The Secret Stamp:** PhonePe and our server share a private secret code (called a `SALT_KEY`) that is never revealed to the public or sent to any browser.
2. **The Digital Signature:** When PhonePe tells our server that a payment succeeded, it creates a unique digital stamp using SHA-256 mathematical encryption that combines the payment details with the secret key.
3. **Verification:** When our server receives the message, it runs the exact same formula. If the calculated stamp matches the stamp on the message, the server knows:
4. The message 100% came from PhonePe's official bank servers.
5. Nobody intercepted or altered the amount along the way.
6. **No Trust in the Phone:** The student's phone browser is never trusted to confirm a payment. Only PhonePe's server-to-server webhook can credit money to a wallet.